



Altruistic punishment in intergroup context

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Abstract

Experimental studies in the literature have shown that individuals are willing to sacrifice their own resources to punish free-riders even if there is no possibility to get reward for this behavior. The aim of the current study is to examine whether individuals have preferences to punish violators when the punishment is costly for themselves and to test whether this preference is affected by social categorization (i.e., ingroup-outgroup settings). For these aims we conducted two experiments. In both studies the Third-Party Punishment Game was used. Study 1 in which a 2 (distributor's identity: ingroup vs. outgroup) x 2 (recipient's identity: ingroup vs. outgroup) x 2 (fairness: equal vs. unequal) within-subjects design was used consists of 38 participants and Study 2 in which a mixed design was used consists of 174 participants. It was found in both studies that participants spent higher amounts to punish unfair players than fair players. Additionally, in Study 2 participants were found to prefer to altruistically punish the perpetrator more harshly when the victim was an ingroup member. The findings suggest that individuals attempt to maintain cooperative behavior norms by engaging in costly punishment of selfish individuals, regardless of their group membership. However, the stricter reactions to unfairness directed to an ingroup member show that participants favor ingroup members over outgroup members in line with the social identity literature.

Keywords Costly punishment · Norm enforcement · Social preferences · Social categorization · Third party

Introduction

Human beings have a tendency to punish those who violate the norms of cooperation and fairness. According to evolutionary theorists (Fehr & Fischbacher, 2004; Tomasello, 2016), people prefer to interfere in unfair situations even if they are third-parties, because this behavior serves to maintain the cooperation within the group. It has been shown that unaffected third-parties prefer to punish the violators at cost rather than not intervene (Shinada et al., 2004).

This tendency is conceptualized as “altruistic punishment” (Fehr & Gächter, 2002). The aim of the current study was to answer the question of whether third-parties prefer to punish those violating the norm of fairness at the cost, as suggested by evolutionary theorists. According to distributive justice literature there are three different criteria to evaluate whether a situation is fair: *need*, *equity* and *equality* (for a further review Colquitt, 2012). In this study “equal sharing” is considered as “fairness” because there is no information about the contribution or need in the game used in this study, so the only distributive justice norm has been the principle of equality rather than equity or need.

Since the individuals interpret their environment through the lenses of their social belongings, we aim to investigate this behavior within the framework of social categorization approach. According to social identity theory (Tajfel et al., 1971), individuals interpret the social world as consisting of ingroups (the social groups in which they are a member) and outgroups (the social groups in which they are not a member). They tend to derive and maintain a positive social identity obtained from the relative status of their ingroup in comparison with the relevant outgroup (Tajfel & Turner,

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1979). By adding this social categorization perspective to the concept of altruistic punishment, we aim to answer the question of whether third-parties' tendency of punishment would be affected by sharing the same social identity as the perpetrator or victim. In terms of the effect of perpetrator's social identity on a third-party's punishment tendency there are two different hypotheses according to regarding literature. *Intergroup bias hypothesis* (Jordan et al., 2014; Yudkin et al., 2016) posits that individuals would punish unjust ingroup members more leniently than unjust outgroup members, because they favor ingroup members against outgroup members and this effect would especially be more strongly if there is a conflict or competition between the ingroup and the outgroup. On the other hand, *black sheep effect hypothesis* (Marques et al., 1988) posits that individuals would punish unjust ingroup members more harshly than unjust outgroup members because they want to regulate the ingroup by protecting group norms. Since the finding that competition between groups increases the intensity of intergroup bias is well established by previous studies (Brewer, 2001; Schiller et al., 2014; Sherif et al., 1961) we attempt to test these two contradictory hypotheses mentioned above in the context of absence of any conflict or explicit competition between the groups. Furthermore, considering the concept of altruistic punishment within the framework of social categorization perspective would be useful to uncover whether this tendency is affected by the social identity of the victim, as well. According to social identity theory individuals empathize more easily with ingroup members than outgroup members since they identify themselves with their groups and as a consequence regard the interests of ingroup members interchangeable with their own (Tajfel & Turner, 1979; Turner et al., 1987: 65). Based on this proposition, it is reasonable to expect individuals to punish the unjust actor more severely when the victim of the act is an ingroup member rather than an outgroup member. We aim to test this hypothesis in the context in which there is no conflict or explicit competition between ingroup and outgroup. Because we avoid to contextual factors which might magnify the intergroup bias, we choose a neutral intergroup context consisting of two groups with no conflict or competition. We attempt to find out whether this natural propensity for punishing norm violators would be affected by only social categorization process with no contextual factors increasing intergroup bias.

The previous research scrutinized cooperative behavior in various intergroup contexts, including minimal groups generated based on an artificial laboratory criterion (Tajfel et al., 1971), real social groups involving competition, such as supporters of a political party or a soccer team (Schiller et al., 2014), real social groups with no intergroup competition and which consisted of randomly assigned members (Goette et al., 2006), and real social groups with

intergroup competition and which consisted of randomly assigned members (Goette et al., 2010). However, the aim of this study is to answer whether willingness to intervene against unfairness is influenced by social categorization in the ingroup and outgroup settings with non-competitive real social groups. This study is expected to contribute the literature on tendencies of third-party punishment in intergroup context consisting of real social groups with no conflict.

Enforcement of the norm of equal sharing

Social norms for cooperative behavior are critical in ensuring the continuity of cooperation through sanctions by third-parties (Boyd & Richerson, 1992; Fehr & Fischbacher, 2004; Fehr & Gächter, 2000, 2002). Particularly, in large-scale groups, the continuity of cooperation is ensured with the help of third-parties punishing the perpetrators even if they do not gain any benefit (Marlowe et al., 2008; Boyd et al., 2003).

Primitive humans acknowledged their interdependence for survival and started functioning together for the common good. Thus, caring for the well-being of partners (sympathy), sharing foods equally with those they hunted together (equal entitlement), and being able to control own impulses and desires for a shared purpose (self-regulation) ensured the cooperation for the continuity of humanity and formed the basis of the moral life of today's communities (Tomasello, 2016). Nevertheless, over time, the groups needed to control free riders jeopardizing the continuation of cooperation by benefitting unilaterally from group cooperation. This necessity initiated the emergence of social norms (Fehr & Fischbacher, 2004; Tomasello, 2016). According to many evolution scholars the evolution of cooperation was possible through the principle of *strong reciprocity* (Falk & Fischbacher, 2006; Fehr et al., 2002), which refers to giving others what they deserve (i.e. rewarding kindness and punishing unkindness). In many cases, third-parties are willing to respond to the behavior they witness at the cost even if they cannot derive any benefit for themselves.

Punishment of perpetrators is often costly for the witnesses as well. Spending time and effort to punish the perpetrator instead of ignoring the violation is a costly behavior for individuals. Yet, it is well-known that individuals have propensity to punish those violating cooperation (Fehr & Fischbacher, 2004). In this sense, just as they do when cooperating, individuals seem not to hesitate to put social preferences before their financial interests while punishing those acting selfishly. Punishing a perpetrator at the cost is driven by moral emotions emerging as a reaction to violation (Folger et al., 2005). An empirical study showed the association between witnessing a selfish action as an unaffected

third-party and feeling anger (Fehr & Gächter, 2002). In that study, the participants reported that they got angry at free riders and expect that other members would be angry with them if they were a free rider. Briefly, the feeling of anger is an important factor which underlies behind altruistic punishment (Fehr & Gächter, 2002; Jordan et al., 2016). It is also claimed that the feeling of guilt, as well as anger, motivates individuals to act toward correcting any negative situation experienced (Nelissen & Zeelenberg, 2009).

Even though third-party punishment within group has been widely researched (Fehr & Fischbacher, 2004; Fehr & Gächter, 2002; McAuliffe et al., 2015), we aim to handle this behavior in the intergroup context by adding social identity perspective to this phenomenon. With this regard, the aim of the study is to investigate the effect of the perpetrator's and the victim's social identity on third-party punishment.

The effect of social category on punishment decision

Perpetrator's identity

In essence, two different hypotheses prevail in the literature regarding the nature of third-party punishment in terms of the effect of perpetrator's identity. According to the group regulation hypothesis (Fehr & Gächter, 2002), the first of these hypotheses, costly punishment for those violating group norms is indeed a group investment. Therefore, third-parties are expected to prefer to punish ingroup members, rather than outgroup members, for violating cooperation norm. In this perspective, individuals care about preserving the norm of cooperation within the group since preferring ingroup members as cooperation partners (Brewer, 2001; Tomasello, 2016). Ingroup members disrupting cooperation are expected to be judged more negatively to maintain the group norms of cooperation, particularly when the victim is also an ingroup member. This is similar to the phenomenon of the "black sheep effect" (Marques et al., 1988), which suggests that individuals judge more harshly the deviant ingroup members than outgroup members engaging in the same behavior, especially when it comes to the group's reputation.

The second hypothesis on altruistic punishment in the intergroup context is the intergroup bias hypothesis (Yudkin et al., 2016). Accordingly, how the perpetrator's action is interpreted is highly affected by intergroup biases, and individuals evaluate ingroup members' violation of fairness norm more leniently because of their positive feelings toward the ingroup (Everett et al., 2015; Otten, 2009). Such a tendency is considered ingroup favoritism from the perspective of social identity theory (Tajfel & Turner, 1979).

The bias of approaching positively toward ingroup members affects the judgment related to punishing them in violation situations, leading to behaving more moderate when punishing ingroup members and being stricter toward outgroup members. In summary, the two hypotheses on the impact of the perpetrator's identity regarding the punishment in the intergroup context suggest different propositions as to whether ingroup or outgroup members should be punished more harshly.

Victim's identity

Against whom the violation is committed is also a contextual variable that may influence the decision of costly punishment for the perpetrator. The previous research consistently showed that individuals tend to punish the perpetrator more harshly when the victim of injustice is an ingroup member than when they are an outgroup member (Rabellino et al., 2016; Goette et al., 2006; Bernhard et al., 2006b). Research on the influence of ingroup or outgroup victimization on altruistic punishment revealed that ingroup victims are more protected than outgroup victims (Schiller et al., 2014), which is called parochial altruism (Bernhard et al., 2006b). The fact that individuals react more harshly to the violation of social rules when the victim is an ingroup member may be empirical evidence of the influence of positive feelings toward the ingroup on altruistic punishment (ingroup love; Brewer, 1999). Besides, the social identity theory considers parochial altruism an ingroup favoritism (Everett et al., 2015).

Aim and scope of the research

The present research was carried out to test whether witnesses altruistically prefer to punish a perpetrator of the norm of equal sharing. To put it another way, we sought an answer to whether people acting selfishly are punished by third parties at the expense of its cost. In this context, we investigated whether there are preferences for altruistic punishment, as indicated by evolutionary theories, and whether these preferences differ in terms of sharing the same social identity with the perpetrator and the victim (ingroup or outgroup member) in the intergroup context.

Research hypotheses and questions

We tested the following hypotheses in two studies:

1. Third parties would punish actors who allocated the resource unequally more harshly than those allocated it equally.

The theorists adopting the principle of strong reciprocity assert that individuals tend to reciprocate good and bad

behavior, regardless of the future benefit and even when they will not encounter that person again (Fehr et al., 2002). The literature on altruistic punishment often proposes that perpetrators adopting a selfish strategy are punished in proportion to their level of selfishness. Namely, actors acting more selfishly are punished more harshly (Fehr & Gächter, 2002). Therefore, we expected that the punishment for unequal allocation would be greater than that for equal allocation.

2. Third parties would feel more anger in conditions of unequal allocations than in those of equals.

Studies examining altruistic punishment as a way of enforcing the fairness norm (Fehr & Gächter, 2002; Shinada et al., 2004) concluded that witnesses experience anger and loss of trust and perceive unfairness and that such negative emotions and judgments predict the punishment behavior. Hence, we expected that our participants would experience more anger in conditions with unequal allocation than those with equal allocation.

3. Third parties would punish the perpetrator more harshly when the victim is an ingroup than an outgroup member.

Research examining the effect of victim's identity on punishment preferences suggested that the amount of punishment given in a case in which an ingroup member is victimized is higher than an outgroup member is victimized (Bernhard et al., 2006a; Rabellino et al., 2016). Thus, the present study hypothesized that protecting the victim would be influenced by ingroup favoritism, which would be in favor of the victims that are ingroup members.

4. Third parties would feel more anger when the victim is an ingroup member than an outgroup member.

According to the social identity theory, when the victim is an ingroup member, witnesses demonstrate similar emotional reactions to the victim, even though the negative behavior will not affect their own outcomes (Dumont et al., 2003). As stated before, it is known that anger motivates third parties to punish the violators of fairness norm (Fehr & Gächter, 2002). Empirical evidence previously showed that witnessing the victimization of an ingroup member leads to greater anger than seeing the victimization of an outgroup member (Yzerbyt et al., 2003). In this study, it was expected that witnesses would feel more anger when the victim is an ingroup rather than an outgroup member.

We also attempted to answer the questions below:

1. Is there a significant difference between the amounts of punishment for an ingroup violator and an outgroup violator?

In the relevant literature on the impact of sharing the same social identity with the violator on punishment, some of the studies found that ingroup members are punished more, supporting the group regulation hypothesis (Mendoza et al., 2014; Shinada et al., 2004), while some others revealed

that outgroup members are punished more, advocating the intergroup bias hypothesis (Yudkin et al., 2016). It was also found that there is no significant difference between the punishment for an ingroup violator and an outgroup violator (Goette et al., 2006; Lu & McKeown, 2018). In this sense, the question of who (an ingroup or outgroup member) is punished more harshly in the violation of fairness norm still remains to be answered. Therefore, we sought an answer to whether altruistic punishment preferences are affected by social categorization when there is no intergroup conflict.

Study 1

Materials and methods

Participants

We remotely collected the data from undergraduate psychology students at Mersin University in Turkey. The data of participants who reported not paying attention to the stimuli on the post experiment survey were excluded. Overall, the sample consisted of a total of 38 participants, 20 females and 18 males, with a mean age of 20.68 years ($SD=1.16$; age range = 18–23 years).

Data collection tools

Altruistic punishment: We assessed altruistic punishment using the Third-Party Punishment Game (Fehr & Fischbacher, 2004). The game, played between three participants with different roles (player A, player B, and player C), measures the reaction of the observer to punish the violator when s/he witnesses a type of interaction between the other two players that involves a violation of equal share. When it comes to the rules of the game, player A is provided an endowment of which s/he can share any portion with player B. How much endowment player B will win in the game is entirely up to player A's decision. Player C, on the other hand, is given an endowment that s/he can either spend all or part of it on punishing player A or keep it all after recognizing A's decision. Three times the portion that player C spent to impose sanctions on player A is deleted from player A's endowment. In this research, we allocated 40 units of the endowment to player A and 20 units of the endowment to player C. While explaining the capacity of player C, we preferred the term "deletion of points" instead of "punishment" to avoid directing the participants. Moreover, the participants were told that the portion remaining in their budget would be converted into Turkish Lira (TRY) at the end of the experiment and paid to them (maximum TRY 10) at the end of the experiment. We kept ambiguous the

exchange rate of money unit and Turkish Lira. In addition, we followed the method by Fehr and Gächter (2002) and eliminated direct or indirect reciprocation expectations to measure altruistic preferences. Accordingly, the participants were instructed that they would encounter different players in each round and never play with the same player again.

The game consists of 20 rounds (16 unfair allocations, 4 equal allocations). Each amount of unfair allocation flashed as a stimulus always varies because seeing the same stimulus will likely raise doubts about the reality of the experiment. Strobel et al. (2011) found that the participants perceived a portion of 0–6 units out of a 20-unit endowment as unfair, a share of 7 units as neutral, and a portion of 8–10 units as fair. In this research, we carried a-out a pilot study employing a different sample to identify how many portions out of 40 units of the endowment are perceived as equal or unfair. The findings revealed that the portions of 1, 2, 3, and 4 units of the endowment out of 40 were perceived equally unfair. Thus, the participants in the main study were shown these four different portions in the unfair condition, and we averaged the punishment responses to each of them. In the fair condition, participants were shown 20 money units of the endowment out of 20. The participants were shown each portion in random order. Overall, we calculated Cronbach's alpha coefficients to be 0.96 for 16 rounds of unfair allocation, and 0.92 for 4 rounds of equal allocation.

We also performed a pilot study to uncover what department the participants perceived as an outgroup at the optimum level and a conflicting group at the minimum level. The findings yielded the Department of English Language and Literature (ELL) to be the most suitable outgroup among others (Sociology, GPC (guidance and psychological counseling), and Mathematics Teaching). During the game, while a psychology student was shown as the ingroup stimulus, the outgroup stimulus was an ELL student. All participants were psychology students and were made to believe that they were randomly assigned to play as a Player C. In fact, they were playing with two simulated Players. They saw 4 different scenarios depending on the experimental conditions regarding the group memberships. An ingroup member (Psychology student) shares the resource with an ingroup member (Psychology student). An ingroup member (Psychology student) shares the resource with an outgroup member (ELL student). An outgroup member (ELL student) shares the resource with an ingroup member (Psychology student). An outgroup member (ELL student) shares the resource with an outgroup member (ELL student).

A round in the game consists of two consecutive screens. The instruction on the first screen informs the participant to wait for the new player's decision. Then, an instruction pops up on this screen: "New players have been selected. Now, player A is expected to tell B what portion of the 40-unit

endowment s/he will transfer to him/her." The first screen disappears after 7 s, and the second screen displays what departments players A and B are enrolled in and what portion of the endowment player A has shared with player B. To measure the altruistic punishment responses of the participants, the second screen pops up a directive: "What portion of your endowment do you want to spend to deduct from the endowment of player A? (3 times the portion selected will be deleted from player A's endowment)." Thus, the participant is asked to choose a number between 0 and 20. Figure 1 presents the snapshots of the consecutive screens of the third-party punishment game used in the present study.

Emotional responses toward the event witnessed: Following the game, we reminded the participants about the stimulus in each experimental condition and asked them to rate their emotions (anger, feeling good, guilt) toward player A's decision on a scale ranging from 1 (not at all) to 5 (completely). Measurement of emotions after assessing punishment was frequently used in the previous research (Karagonlar & Kuhlman, 2013; Pillutla & Murnighan, 1996; Srivastava et al., 2009) since eliminating the possibility of participants' changing their punishment responses based on their previous emotional responses. For the same reason, we measured emotions after assessing altruistic punishment.

Procedure

The Social and Human Sciences Ethics Committee of Mersin University granted the ethical approval to our study (dated 12/31/2018 and numbered 017). We adapted the experiment file, prepared on the E-Prime 3.0 software (2016, Psychology Software Tools) for remote data collection, through the E-Prime Go extension and invited the participants to play a 3-player online game through student groups. Voluntary students filled a google form indicating their e-mail addresses and the time they will be available at which they would fully be quiet and alone. Then we sent an electronic link of the experiment to their e-mail addresses on the day and time they specified. Following a one-round trial game, they started the actual game with 20 rounds. The participants were told to play with two other real players (player A and player B), yet, in reality, all the participants assigned for the role of player C and played against two simulated characters.

At the end of the experiment, we measured the participants' emotional responses and collected their demographic information. Also, we sought their estimations about the subject and purpose of the experiment through their responses in the feedback form. Finally, although we announced that their remaining endowment at the end of the 3-week data collection would be paid to them, each participant was paid

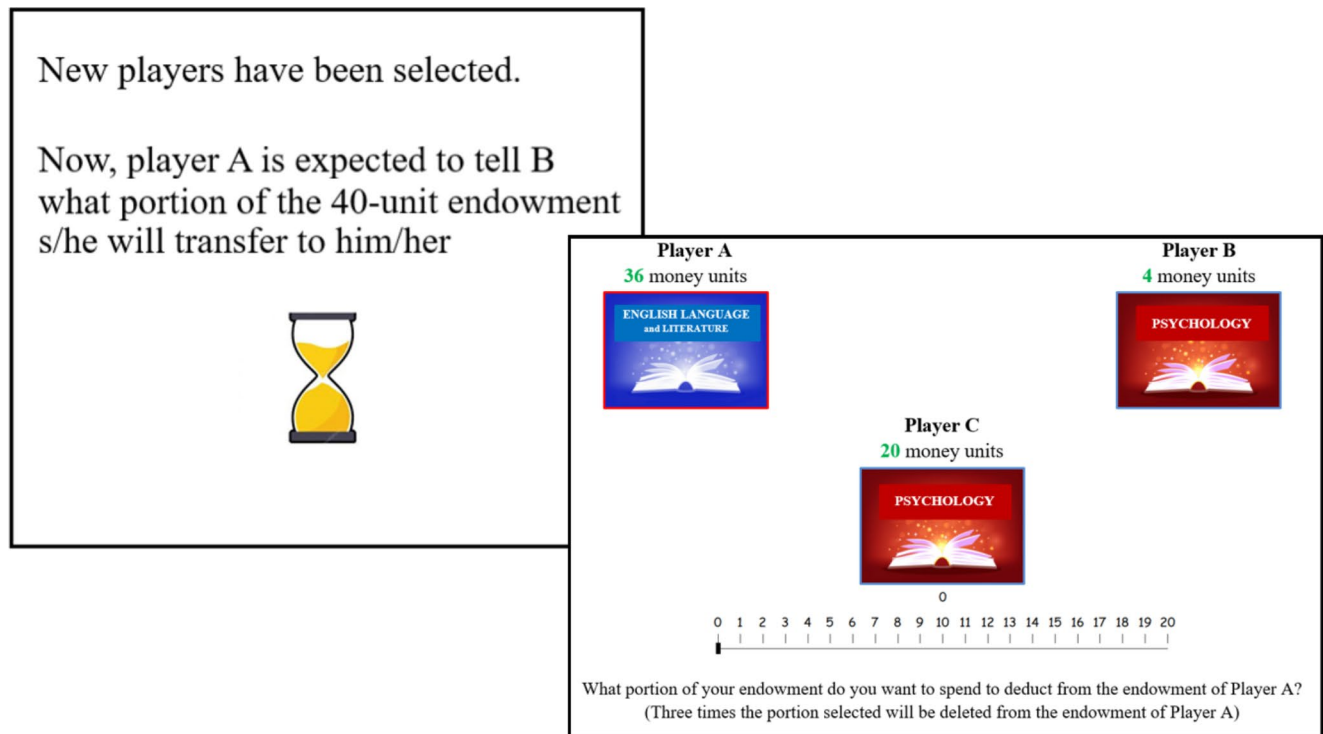


Fig. 1 Overview of the one trial run of the third-party punishment game used in the present study

TRY 10 via internet banking regardless of their remaining balance. The experiment took about 50 min, and the data were collected between September and October 2020.

Data analyses

In the present study, we attempted to test whether altruistic punishment behavior (first purpose) and emotional responses (second purpose) significantly differ by recipient's identity, distributor's identity, and fairness of the allocation. Thus, we performed two separate 2 (recipient's identity: ingroup vs. outgroup) $\times$ 2 (distributor's identity: ingroup vs. outgroup) $\times$ 2 (fairness: unfair vs. equal) repeated measures ANOVA. IBM SPSS Statistics Version 20 was used for all analyses. We initially included the gender as a variable in the analyses and the results revealed that the main effect of the gender was not significant according to the result of the 2 (distributor's identity) $\times$ 2 (recipient's identity) $\times$ 2 (fairness) $\times$ 2 (gender) mixed design ANOVA ($F(1,36)=0.47$, $p=.5$, partial $\eta^2=0.01$). In addition, none of the interaction effects of gender was significant. Thus, we excluded it from the subsequent analyses since its effect is not significant.

Results and discussion

Results of altruistic punishment

Fairness. The results revealed that only the main effect of the fairness was significant (Wilk's $\lambda=0.29$, $F(1,37)=93.05$, $p=.00$, partial $\eta^2=0.72$). Accordingly, the amount of punishment for the unfair conditions ($M=9.15$, $SD=2.99$) was found to be significantly higher than that for the equal conditions ($M=3.03$, $SD=3.57$). Yet, the main effects of distributor's identity and recipient's identity were not significant. In addition, we could not find any significant interaction effect.

Results of emotional responses

Fairness. The analysis revealed a significant main effect of fairness (Wilk's $\lambda=0.22$, $F(3,35)=40.70$, $p=.00$, partial $\eta^2=0.77$). Accordingly, anger ($F(1,37)=66.79$, $p=.00$, partial $\eta^2=0.64$), guilt ($F(1,37)=15.56$, $p=.00$, partial $\eta^2=0.30$), and feeling good ($F(1,37)=121.11$, $p=.00$, partial $\eta^2=0.77$) differed significantly by fairness of allocation. We concluded that the participants felt more anger ($M=2.97$, $SD=1.18$) and guilt ($M=1.88$, $SD=1.04$) in the unfair allocations when compared to the fair allocations ($M=1.28$, $SD=0.53$ and $M=1.20$, $SD=0.46$, respectively). In addition, the participants reported feeling better when

witnessing the equal allocations ($M=1.51$, $SD=0.54$) than encountering the unfair allocations ($M=3.62$, $SD=0.94$).

Distributor's identity. We also found a significant main effect of distributor's identity (Wilk's $\lambda=0.80$, $F(3,35)=2.98$, $p=.044$, partial $\eta^2=0.20$). Anger ($F(1,37)=4.78$, $p=.035$, partial $\eta^2=0.11$) and guilt ($F(1,37)=6.18$, $p=.018$, partial $\eta^2=0.14$) significantly differed by distributor's identity. In this regard, the participants reported significantly more intense anger ($M=2.19$, $SD=0.70$) and guilt ($M=1.62$, $SD=0.68$) when the distributor was an outgroup member than when s/he was an ingroup member ($M=2.06$, $SD=0.67$ and $M=1.46$, $SD=0.60$, respectively).

Overall, we concluded that the distributor engaging in unequal allocations was given more altruistic punishment and aroused more anger, more guilt and less positive feeling than the one who engaging in fair allocations. First and second hypotheses of this study were supported. This finding is in line with the concept of altruistic punishment (Fehr & Gächter, 2002). However, according to the findings in terms of the nature of this tendency for altruistic punishment in ingroup-outgroup settings, this tendency did not differ by the context of ingroup/outgroup. Therefore, third and fourth hypotheses were not supported. The within-group design may have led the participants to notice that the distributor's and the recipient's identities changed systematically between the conditions and to show socially acceptable reactions. Another reason why we could not find the group effect may be because the participants played the game for many rounds in a row and may not have paid attention to the other players' group affiliation. Thus, we carried out another study where the social identities of the distributor and the recipient were designed as between-group variables.

Study 2

Materials and methods

Participants

Undergraduate psychology students at several universities remotely participated in the second study. We excluded the data of participants who reported not paying attention to the stimuli on the post experiment survey. The sample consisted of 174 participants, 139 females and 35 males, with a mean age of 21.16 years ($SD=2.82$; age range = 18–45 years).

Data collection tools

We adopted the exact instructions as in the Third-Party Punishment Game in the first study to measure altruistic punishment. However, unlike the first study, the endowment

allocated to the observers was defined as “raffle tickets,” which would increase their chances of winning money. At the beginning of the experiment, the participants were instructed that 5 participants would win TRY 50 each in the raffle, 20 tickets would be allocated to them for this raffle in each round, and the more tickets they would hold at the end of the game, the higher chance they would have to win the money. In short, deleting a portion of the endowment from player A reduced the likelihood of the actual participants (player C) winning the money. As in the first study, the participants would lose 1/3 of their endowment while imposing sanctions, meaning that spending a ticket would deduct a portion of TRY 3 from player A's budget. Since there was a raffle in the second experiment, to prevent the behavior of punishing player A altruistically from being confused with the behavior of deleting money from player A's endowment to have more chance in the raffle, we warned the participants that the exchange of endowment between players A and B would happen independently of the raffle. The eight-round game consists of 4 rounds of unfair (TRY 1, 2, 3, and 4) and 4 rounds of equal (TRY 20TL) allocation, and five different allocations (TRY 1, 2, 3, 4, and 20) are displayed in random order in the game. For reliability concerns, we calculated Cronbach's alpha coefficients to be 0.86 for 4 rounds of unfair allocation, and 0.83 for 4 rounds of equal allocation. In addition, we used the demographic information form and the questions for emotional responses as in the first study.

Procedure

Some faculty members were asked to announce the call for the experiment in their classes. In this experiment, we followed the same procedure as in the first study. Voluntary students filled a google form and indicated their e-mail addresses and the time they will be available. Then we sent an electronic link of the experiment to their e-mail addresses on the day and time they specified. As in the first study, the participants were employed in the experiment through their computers using the experiment file created with E-prime. The experiment took about 30 min, and the data were collected between December 2020 and May 2021. At the end of the entire procedure, the winners were paid TRY 50 each.

Data analyses

To test whether altruistic punishment differs by *recipient's identity*, *distributor's identity*, and *fairness of allocation*, we performed 2 (recipient's identity: ingroup or outgroup) $\times$ 2 (distributor's identity: ingroup or outgroup) $\times$ 2 (fairness of allocations: unfair or equal) ANCOVA (repeated measures on the last factor). Since the gender variable was not evenly distributed, we statistically controlled the

effect of gender. Thus, we assigned recipient's and distributor's identities as the between-group variables, fairness of allocation as the within-group variable, and gender as the control (covariate) variable. All relevant data (Study 1 and Study 2) are available in the Open Science Framework at DOI <https://doi.org/10.17605/OSF.IO/Z2R3J> (https://osf.io/z2r3j/?view_only=ce5dd3f818cd46958a433975ba78ed11).

Results and discussion

Results of altruistic punishment

Fairness. Our findings showed that the main effect of fairness of allocation was significant (Wilk's $\lambda=0.96$, $F(1,169)=7.26$, $p=.008$, partial $\eta^2=0.04$). Thus, the amount of punishment to the unfair conditions ($M=6.04$, $SD=3.90$) was significantly higher than that to equal allocations ($M=1.76$, $SD=2.63$). Yet, we could not find significant main effects of distributor's and recipient's identities.

Recipient's identity $\times$ Fairness. On the other hand, only recipient's identity and fairness of allocation yielded a significant interaction effect (Wilk's $\lambda=0.96$, $F(1,169)=7.23$, $p=.008$, partial $\eta^2=0.04$). We then performed pairwise comparisons to identify the source of the significant difference between the groups. The results yielded that the amount of punishment was significantly higher in the case of unfair allocations to the ingroup recipient ($M=6.69$, $SD=4.11$) than to the outgroup recipient ($M=5.43$, $SD=3.60$). In the equal allocations, whether the recipient was an ingroup ($M=1.58$, $SD=2.51$) or an outgroup member ($M=1.93$, $SD=2.76$) did not yield a significant difference between the punishment scores (Fig. 2).

Results of emotional responses

Controlling the effect of gender, we performed a 2 (recipient's identity: ingroup vs. outgroup) $\times$ 2 (distributor's identity: ingroup vs. outgroup) $\times$ 2 (fairness of allocation: unfair vs. equal) MANCOVA to test whether emotional responses differ by recipient's identity, distributor's identity, and fairness of allocation.

Fairness. The findings demonstrated a significant main effect fairness of allocation (Wilk's $\lambda=0.75$, $F(3.167)=18.54$, $p=.00$, partial $\eta^2=0.25$). Accordingly, anger ($F(1,169)=19.18$, $p=.00$, partial $\eta^2=0.10$) and feeling good ($F(1,169)=53.27$, $p=.00$, partial $\eta^2=.24$) differed by fairness of allocation. The participants experienced more anger in the unfair allocations ($M=3.28$, $SD=1.31$) when compared to the equal allocations ($M=1.32$, $SD=0.81$). Similarly, the participant reported feeling better in the equal allocations ($M=3.79$, $SD=1.10$) than the unfair allocations ($M=1.48$, $SD=0.87$). However, we could not reach a significant difference in experiencing guilt by fairness of allocation.

Recipient's identity $\times$ Fairness. The results also exhibited that only the interaction effect of "recipient's identity $\times$ fairness" was significant (Wilk's $\lambda=0.94$, $F(3.167)=3.24$, $p=.023$, partial $\eta^2=0.06$). According to the relevant F tests, this interaction effect was rather close to "significant" in anger ($F(1,169)=3.79$, $p=.053$, partial $\eta^2=0.02$) but not in other emotions. Still, we performed pairwise comparisons and found that the anger level was significantly higher when an ingroup recipient experienced unfairness ($M=3.50$, $SD=1.21$) than when it was so for an outgroup recipient ($M=3.06$, $SD=1.38$). Figure 3 illustrates a graph for the mean anger by fairness of allocation (equal vs. unequal) and recipient's (player B) identity (ingroup vs. outgroup).

Fig. 2 Expenditures to punish distributor by recipient's identity and fairness of division. Responses were between 0 and 20 of tickets to punish Player A

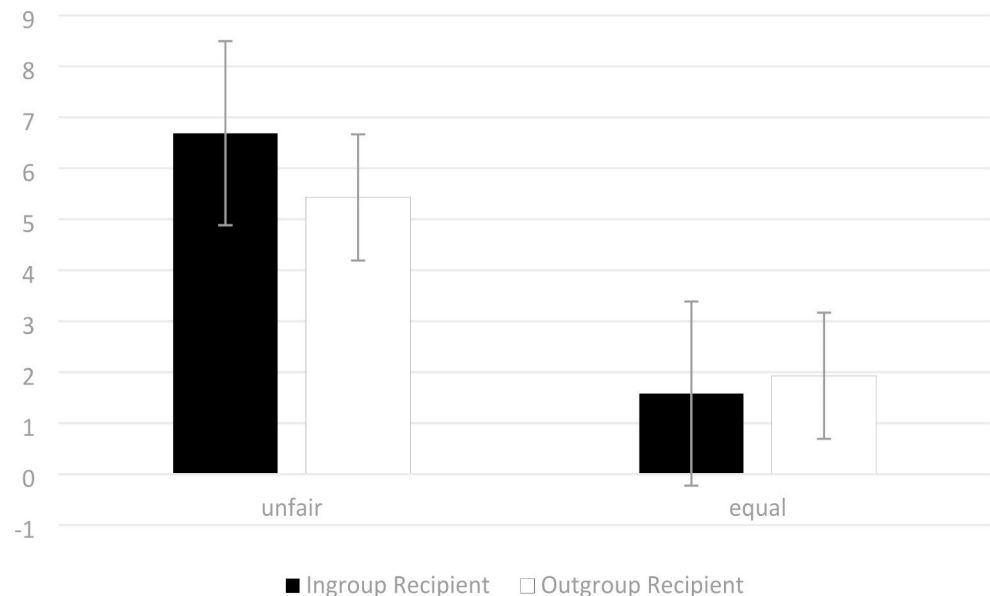
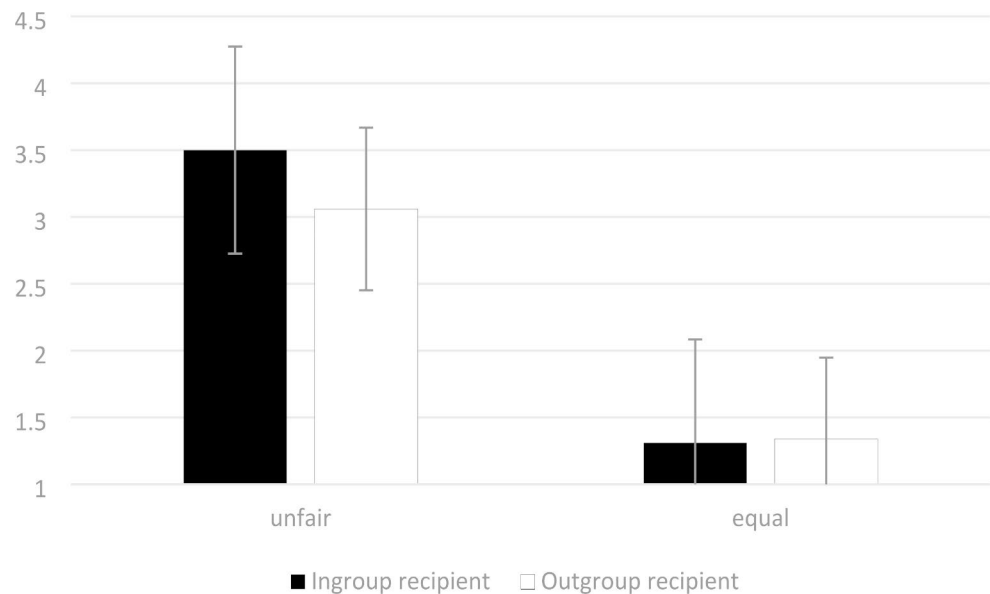


Fig. 3 Anger at distributor by recipient's identity and fairness of division. Responses were between 1 (not at all) to 5 (completely) for anger



The findings of the Study 2 showed that unfair players were given more altruistic punishment and aroused more anger and less positive feeling than fair players by supporting the first and second hypotheses as in the Study 1. This finding can be considered as an evidence for altruistic punishment (Fehr & Gächter, 2002). Additionally, according to the findings in terms of the nature of this tendency in ingroup-outgroup settings, this tendency was affected by the social identity of the victim. When the victim was an ingroup member participants punished unfair players more harshly and reported more anger at unfairness. This finding supported the third and fourth hypotheses of the study which is in line with the social identity theory (Tajfel et al., 1971) suggesting that people favor ingroup members over outgroup members. However, we found no effect of the perpetrator's social identity on punishment decisions.

General discussion and conclusion

The ultimate aim of the present study was to investigate whether the third parties preferred a costly punishment for the perpetrator for violating a social norm, as advocated in the strong reciprocity principle, and whether the group regulation hypothesis or the intergroup bias hypothesis is prevailing in punishing the perpetrator in the conflict-free intergroup context. The findings of both experiments uncovered that the unfair allocations were punished more harshly than the equal allocations within the conditions set in the study and that the unfair allocations led to more anger. In addition, according to the findings of Study 1 unfair allocations led to less positive feeling than the equal allocations.

The strong reciprocity principle proposes that individuals sustain the norm of cooperation by punishing those violating

the norm, even if it is costly for them (Fehr et al., 2002). In our study, the participants preferred the costly punishment for the perpetrator in one-shot interactions that did not offer any profit, which robustly endorses the strong reciprocity principle (Fehr & Gächter, 2000; Gintis, 2000). In addition, our findings are consistent with the previous research concluding that the feeling of anger when witnessing unfairness motivates one to engage in altruistic punishment behavior to impose the norm of cooperation (Fehr & Gächter, 2002; Folger et al., 2005; Jordan et al., 2016).

Considering altruistic punishment behavior in the intergroup context, we found that perpetrator's identity had no significant effect on altruistic punishment behavior in both experiments. The reason why we could not find a group effect in altruistic punishment preferences in terms of perpetrator's identity in line with the intergroup bias hypothesis, may be that the acts of outgroup members were not perceived as symbolic threats to social identity. According to Wenzel (2009: 64), it becomes disturbing for individuals when perceived unfair resource allocation involves a symbolic threat to social identity. Thus, individuals are more likely to show punitive reactions to outgroup members when they attribute their unfair acts to the intentions of posing superiority to or underestimating the status/power of the ingroup, that is, when ingroup members think that outgroup members disrespect their social identity (Otten, 2009: 164–165; Wenzel, 2009). Yet, the absence of any threat to social identity is likely to lead individuals to be unwilling to distribute negative resources to outgroup members (Brewer, 2001; Mummendey & Otten, 1998). In our study, since the intergroup context was composed of non-competitive and relatively unbiased groups, the participants may not have perceived the violations of outgroup members as a threat to

their social identity. Therefore, they may not have punished the outgroup violator more than the ingroup violator.

Although our finding showing the no significant effect of perpetrator's identity on altruistic punishment did not support the intergroup bias hypothesis, it does not contradict the group regulation hypothesis either (Fehr & Gächter, 2002; Mendoza et al., 2014; Shinada et al., 2004). The hypothesis assumes that third parties are expected to spend more on punishing the ingroup member violator for maintaining the norm of cooperation since altruistic punishment is often said to serve to preserve the ingroup norm of cooperation. In both experiments, our participants may have perceived complying with the norm of equal sharing as the responsibility of all, not just a few ingroup members. The participants' preference to punish altruistically those violating the equal sharing norm, but these preferences' not being affected by the perpetrator's group affiliation, may imply that the participants considered the norm of fairness a rule that humanity, a more inclusive superordinate group, should abide by. An experiment by Schmidt et al. (2012) revealed that the participating children imposed the rules of the game (conventional norms) only on ingroup members and moral norms on both ingroup and outgroup members. Scholars reported that young children (from 3 years) have preferences for playing games with ingroup members and, therefore, intervene more when a rule is violated by ingroup members. However, they do not distinguish between ingroup and outgroup members in intervening in a moral norm violation; children equally protest both ingroup and outgroup perpetrators. Our finding that sharing a common group affiliation with those violating the norm of fairness did not affect altruistic punishment preferences is consistent with the finding of Schmidt et al. (2012) that moral norm violations apply to everyone.

Overall, we concluded that the third parties imposed the norm of fairness equally on both ingroup and outgroup violators in the absence of intergroup competition and conflict. The same finding in the first (distributor's and recipient's social identities were considered within-group variables) and the second studies (these variables were considered as between-group variables) endorses previous research showing that altruistic punishment preferences are not influenced by the violator's group affiliation in conflict-free intergroup contexts (Goette et al., 2006, 2010).

Considering altruistic punishment preferences by victim's identity in the intergroup context, we discovered it not to affect altruistic punishment preferences in the first study utilizing a within-group design. However, this effect was found to be significant in the second study in which the violator's and the victim's social identities were adopted as between-group variables. The second study resulted in that the third parties were more prone to protect ingroup victims, as expected. In this regard, the participants spent more on

punishing the violator for the sake of the ingroup victims than the outgroup victims. The previous research suggested that altruistic punishment was often used to protect ingroup members in the Third-Party Punishment Game (Goette et al., 2006; Bernhard et al., 2006b). When it comes to emotional responses in the second study, we found that the participants were angrier at victimizing the ingroup members than the outgroup members, consistent with their altruistic punishing behavior, which is also compatible with the findings of the previous research (Yzerbyt et al., 2003). The fact that the participants reacted more sensitively to the victimization of an ingroup member is also consistent with both the Empathy-Altruism Hypothesis and the implications of the Social Identity Theory. The Empathy-Altruism Hypothesis states that empathic concern underlies altruistic helping behavior, and individuals exhibit ingroup favoritism in empathy (Cikara et al., 2014; Hein et al., 2010). In other words, one may empathize more easily with ingroup members; thus, the use of more punishment when ingroup members are wronged can be explained by ingroup favoritism in empathy. The finding of the second experiment on ingroup favoritism by victim's identity also overlaps the previous research (Batson & Ahmad, 2009; Hein et al., 2010) arguing that empathy is directed more toward ingroup members. In addition, identification with a group creates social preferences based on the interests of ingroup members (Simpson, 2006; Tomasello, 2016). According to Social Identity Theory, one begins to perceive themselves as a prototype of the group when defining themselves as a member of a group. Hence, they see the interests of the group and ingroup members as equivalent to their own interests (Everett et al., 2015; Tajfel & Turner, 1979; Turner et al., 1987: 65; Turner & Reynolds, 2012: 401–406). Since individuals define themselves considering their group membership, they follow the group goals as their own (Tajfel & Turner, 1979), which forms the basis of behavior offered to the benefit of group members, such as collective action and cooperation within the group and with ingroup members. For this reason, it is not surprising that individuals will come to their aid when ingroup members are wronged. The fact that ingroup bias was not found in terms of whom the violation was committed in the first experiment, in which within-group design was used, might be related to the use of impression management strategy of participants not to seem biased.

Limitations, recommendations and conclusion

This study is not free of a few limitations. Above all, we had to collect the data in the 2020–2021 academic year when educational institutions shifted to distance education due to the COVID-19 pandemic. Since the participants were involved in the research at their homes, not in a laboratory

setting where external stimuli can be controlled, the control of their physical environments was left to their responsibility throughout the data collection process. Hence, not being able to collect the data in a laboratory environment is a significant limitation of this study. However, we attempted to loosen such a limitation as much as possible throughout the study. The announcement for the research emphasized the need for the participants to provide a private space in a quiet environment during their participation in the research (for approximately one hour). Therefore, we initiated the data collection procedure at the most suitable date and time they determined to create such an environment for themselves. Finally, we did not include the data of those reporting not being able to focus their attention on the game or understand the game instructions or questions.

Secondly, we took the measurements for emotional responses after assessing punishment behavior. We thought an opposite procedure would disrupt the game flow and reduce its credibility. Supposing that the participants had been asked to assess their feelings about the stimuli prior to giving a behavioral response in the game, their focus on their emotions might have confounded their altruistic punishment responses. For these reasons, we thought that measuring emotional responses before behavior would be more disadvantageous than the method we utilized in this study. Considering this limitation, the previous research deploying the functional magnetic resonance imaging (fMRI) technique to investigate emotional reactions during altruistic punishment behavior (Sanfey et al., 2003) or testing the relationship between emotional responses and punishment behavior through regression analysis (Jordan et al., 2016; Mendoza et al., 2014) concluded that negative affect, particularly anger, becomes an important source of motivation explaining individuals' spending their endowment on punishing the perpetrator. Our findings are consistent with the results of previous studies using different techniques to measure emotional responses.

Third limitation of this study is the possibility of the maximum punishment amounts of fair and unfair conditions were not equal if participants calculated the deduction points necessary to reduce the distributor's endowment to zero and decided accordingly. In the case when the distributor kept him/herself 36 points by acting unfairly, the maximum amount for punishment might be 12 points to reduce the distributor's endowment to zero according to the rules of the game (the tripled amount would be reduced from the distributor's payoff). On the other hand, in the case when the distributor kept him/herself 20 points by acting fairly, the maximum punishment cost might be 7 points. In this study nearly 75% of the participants have never selected the punishment point above 3 across fair conditions and the vast majority of the participants did not go above the maximum

value. However, this should be considered as a methodological limitation and this limitation should be overcome by next studies. Another limitation of the present research originated from the fact that the endowment allocated to the participants to assess how they would use it to punish the perpetrator was symbolic "money" that was turned into a real prize later (money unit or raffle ticket), whose amount was low, and that was paid following the game. In this regard, future research is recommended to keep the endowment relatively larger and present the game with real money instead of money unit or raffle tickets. Thus, prospective researchers may pay the money to the participants at the beginning of each round and create situations where participants will instantly lose their money while applying altruistic punishment to the violator.

The final limitation of this study is that the participants were not offered more behavioral options except from punishing the perpetrator. Although the participants were told that they could choose "zero" if they did not want to spend their endowment, it is not possible to completely eliminate the explanation that participants behaved in a way that confirms the researcher's expectation. When we checked the money spent by participants across conditions, we saw that some participants spent their money to deduct from the endowment of fair players even if the mean of deduction points in fair conditions were lower than those in unfair conditions. Considering this finding showing that some participants spent their money to deduct from the endowment of fair players, the necessity of giving different options to react to participants becomes more prominent. In order to overcome the limitation of this study related with demand characteristics, future research may offer participants the options such as not reacting or compensating the victim.

We realized that the participants were willing to spend their shares on punishing the violator. Since the relevant research often scrutinizes this subject through behavioral economic games in an experimental environment, it is still needed to elaborate on whether the behavior above is observed in daily life or under what kind of conditions it arises. Henrich et al. (2006) not only reported that altruistic punishment behavior is observed in many cultures but also argued that it is more prevalent in communities where cooperation and the norm of fairness are highly adopted. Moreover, the literature consistently showed that individual differences are also among the factors moderating altruistic punishment behavior in terms of one's predisposition to cooperation (Shinada et al., 2004) or other-regarding justice sensitivity (Lotz et al., 2011). Hence, further research may be interested in exploring the prevalence of altruistic punishment behavior outside the laboratory game context and uncovering individual and contextual factors that may be associated with this behavior.

In conclusion, preferences for punishing the person violating the norm of fairness in the game were not influenced by sharing the same social identity with the violator in the case of non-competitive intergroup relationships. In other words, the participants did not punish outgroup violators more than ingroup violators. However, when it comes to the victim's social identity, we noticed that the participants were willing to spend more of their endowment on punishing the violator for the sake of the ingroup victims. That is, the third parties' behavior to punish the person violating the norm of fairness appeared in favor of ingroup victims. In this respect, our findings overlap the literature proposing that ingroup favoritism between the non-competing/conflicting groups is not fed by negative attitudes toward the outgroup but by ingroup affection (Brewer, 1999, 2001; Turner & Reynolds, 2004: 266–269).

Author contributions Both authors contributed to the study conception and design. Material preparation and data collection were performed by Özge Ünal analysis was performed by Özge Ünal and Serap Akgün. The first draft of the manuscript was written by Özge Ünal. Serap Akgün commented on previous versions of the manuscript. Both authors read and approved the final manuscript.

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Data Availability All relevant data (Study 1 and Study 2) are available in the Open Science Framework at DOI <https://doi.org/10.17605/OSF.IO/Z2R3J> (https://osf.io/z2r3j/?view_only=ce5dd3f818cd46958a433975ba78ed11).

Declarations

Ethics approval The Social and Human Sciences Ethics Committee of Mersin University granted the ethical approval to our study (dated 12/31/2018 and numbered 017).

Consent to participate Informed Consent was obtained from all individual participants included in the study.

Conflict of interest The authors have no conflict of interest to declare.

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